Bun – The New Runtime On The Block

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JavaScript is the language that we all know and love. Well, some of us dislike it for our own reasons. On one hand, JavaScript is a very flexible and easy to use language with a very large community behind it. You can build anything you want in JavaScript. It may be a complex backend of a website, in-memory databases, really pretty looking frontend or even physics simulations and games! On the other hand, there's so much undefined behaviour in JavaScript that it's easy to get frustrated and drowned in a pile of console errors if you don't know what you're doing. As the famous YouTuber Fireship.io once said "There are a lot of ways to shoot yourself in the foot using JavaScript".

WHAT IS A JAVASCRIPT RUNTIME?

Enough about JavaScript now. Let's talk about JavaScript runtimes. What are JavaScript runtimes? Why do you even need a JavaScript runtime? Let us explain it to you in brief. You see, as the world moved on to better things, people felt the need to run JavaScript outside the browser too. Until then, the only way that you could run and execute JavaScript was inside a browser. Which made JavaScript a client-side scripting language. And it still is a client-side scripting language. But one man by the name of Ryan Dahl thought to himself, "What if we made JavaScript to execute outside the browser too?" and then, NodeJavaScript was born. Node allowed users to run, execute and work with JavaScript outside a browser. Node now made it easier to develop server side applications with JavaScript since none of the JavaScript code could now be seen by the client because it lived on the server. Node took the web development world by storm and well, the rest is history. Now, a runtime is an environment to execute some code. JavaScript now has two runtime environments.

One, inside the browser and the other one outside the browser. Node is a JavaScript runtime environment. Similarly, Deno, which was made by the same guy who created Node, is a JavaScript runtime environment too. The topic of this article – Bun, is a JavaScript environment too. Well, if all of the JavaScript runtime environments can essentially execute the same JavaScript code then why do we need so many of them? What even is Bun? And why are we even covering it with this article? Read on to know.

WHAT IS BUN?

Bun, at its core, is essentially a JavaScript runtime. Just like Node or Deno, it will allow you to run JavaScript outside the browser. You will be able to execute JavaScript files with it and do all the stuff that you would usually do with Node. Okay, that sounds pretty boring to begin with. If it does the same stuff as Node then what's the need for it? Well, just because it is a JavaScript runtime doesn't mean that it has to be similar in architecture as Node or Deno. For comparison, Node and Deno both are built on top of Chrome's V8 engine which is an open source JavaScript engine. But Bun? It uses something entirely different. Bun uses the JavaScriptCore Engine from WebKit. WebKit is Apple's implementation of a JavaScript engine. Just like Chrome uses its V8 engine to power its Chrome browser, Apple uses WebKit to power AppStore, Safari, etc. WebKit, according to Bun, is a little bit faster to start and perform as compared to Chrome's V8 engine. Aside from that, Bun is written in a low level language called Zig. And since Zig is a low level programming language, it provides much more manual control over memory as compared to other languages. That's exactly why Bun is marketing itself to be a lot faster than the other guys in the market.

The best thing about Bun, at least in our opinion, is the native transpilation. Quoting a line from Bun's website, "In bun.js, every file is transpiled. TypeScript & JSX just work". That's a really good thing. Usually, if you wanted to write a Node project in TypeScript, you would have to run two terminals – one which compiles the TypeScript and the other which runs the actual Node app. But now, since everything is transpiled, we don't have to worry about that!

GETTING STARTED WITH BUN!

Unfortunately, Bun is still in beta. That means it hasn't reached its 1.0 major release yet. But that does not mean that we don't get to try this